

Research Internship

IRS Codebook Design under Hardware Considerations

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Research Project

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for

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Intelligent reflecting surfaces (IRSs) can provide an additional propagation path in wireless communication systems. This is especially useful for wireless communication systems employing high carrier frequencies if the Line-of-Sight (LoS) from the transmitter to the receiver suffers from severe attenuation from large objects, such as buildings. IRS comprise a large number of unit cells that can apply a phase shift to the reflected electromagnetic waves. The superposition of the reflections from all unit cells leads to reflection behaviors depending on the phases. Early works on IRS focused on optimizing the phase shifts for perfect channel state information (CSI). Later, however, the problem of channel acquisition was commonly acknowledged and its necessarily large overhead makes such approaches challenging. To avoid resource-intensive channel estimation, the IRS can be configured by selecting predefined codewords from a codebook, which can be done based on measurements of the user's direction. An example for the beam pattern of three different codewords targeting the same direction, but with different beamwidth is shown in Figure 1. In this project, a codebook that is suitable for the near-field shall be derived. In the practical implementation of IRS hardware, the codebooks are limited by the IRS technology, e.g., copper patches adjusted by PIN diodes. Therefore, the limitations and inaccuracies of the hardware shall be considered.

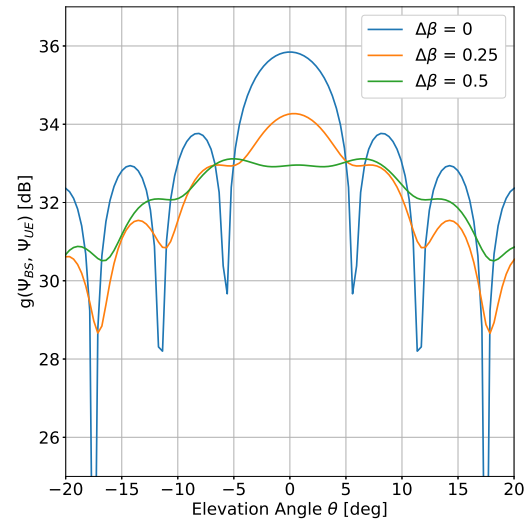


Figure 1: Example for IRS reflection patterns. Here, the parameter $\Delta\beta$ changes the width of the beam.

The main guidelines for the project are as follows:

- Literature review on IRS models and codebook designs.
- Comparison of existing codebooks under hardware impairments and near-field assumption.
- Design of a robust codebook suitable for the near-field under hardware impairments.

PREREQUISITES	Knowledge of digital communication and mathematical optimization.
PROGRAMMING SKILLS	Experience in Python or MATLAB.

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Declaration

To the best of my knowledge and belief this work was prepared without aid from any other sources except where indicated. Any reference to material previously published by any other person has been duly acknowledged. This work contains no material which has been submitted or accepted for the award of any other degree in any institution.

Erlangen, January 11, 2023

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Contents

Title	i
Abstract	vi
Glossary	vii
Abbreviations	vii
Operators	vii
List of Figures	viii
1 Introduction	1
2 Near and Far Field Distances for Extremely Large Antenna Arrays/IRS [1]	4
2.1 Fraunhofer Distance and Fraunhofer Array Distance	5
2.2 Björnson Distance	6
3 System Model: Far-Field Scenario	8
3.1 System Model	8
3.2 Transmission Mode	9
3.3 DFT Phase Shift Design	10
3.4 Linear Phase Shift Design[2]	11
3.5 Quadratic Phase Shift Design[2]	11
4 Uniform 3D Grid Codebook Design for Near Field	13
4.1 System Model: Near-field scenario	13
4.2 Phase Shifts Design using Uniform 3D Grid Space	15
4.3 Results	16
5 Discrete Phase Shift design and Effects of Erroneous Phase Shift Settings	19
5.1 Discrete Phase Shift Model	20
5.2 Effect of Phase Errors and Gaussian Noise on Linear codebook phase shifts	21
5.2.1 Effect of phase errors on IRS beam pattern	22
5.2.2 Impact of Gaussian Noise on Fixed Codebook Phase Shifts . . .	25
6 Conclusion	27
Bibliography	28

Abstract

Intelligent Reflective Surfaces (IRS) is an artificial surface consisting of a continuous or even discrete set of antenna elements with half the wavelength separation between them that can manipulate the impinging electromagnetic waves over the surface when the Line of Sight is considerably blocked. These antenna elements are passive in nature meaning they don't amplify the impinging waves however, just reflect or direct them toward the target using only the necessary power. In this project, we consider the functioning of Intelligent Reflective Surfaces (IRS) in high-frequency regime, meaning considering a model which works in the near field regime as the number of the IRS elements is quite large. We propose our phase shift design for near field scenarios as an extension to the already defined Linear codebook from the literature on far-field regime [2]. We further discuss and define a codebook design scheme to produce discrete phase shifts in our proposed model such that the hardware limitations in terms of phase shift calculations of an IRS surface are met.

Glossary

Abbreviations

IRS	Intelligent Reflective Surfaces
RIS	Reconfigurable Intelligent Surfaces
AoA	Angle of Arrival
AoD	Angle of Departure
BS	Base station
EM	Electromagnetic Waves
mmWave	Millimeter Wave
LoS	Line-of-Sight
MIMO	Multiple Input Multiple Output
ELAA	Extremely Large Antenna Arrays

Operators

$(\cdot)^H$	Matrix Hermitian, Matrix Conjugate Transpose
$\ \cdot\ $	Euclidean Norm

List of Figures

1.1	Example use case scenario of IRS [3]	2
2.1	The phase shift between antenna elements is a result of impinging waves with a delay of $d - d'$ and the phase shift is given by $\frac{2\pi(d - d')}{\lambda}$	6
4.1	The figure shows the IRS lies in the xy plane and the direction of the wave propagation. The point source transmitter lies at a position of $[0, 0, d]$, where d is the distance from the IRS center to the BS [3] . . .	14
4.2	Receivers position with the transmitter in the figure acting as a receiver for our downlink case and IRS is the transmitter which reflects the signal towards the receiver [1]	15
4.3	In Fig (a), IRS response function gain (in dB) vs Distance from the center of the IRS, the Linear codebook for the far-field case using the near-field gain formula [2]. In Fig (b), we use our proposed phase shift design for near-field scenarios and use the near-field gain formula. We plot IRS response gain for various spherical grids, with a constant codebook, and the plot also explains the user position on the spherical grid at radius	18
5.1	IRS reflection gain (in dB) against evaluation angles θ_r for a fixed user at a distance of 30 meters, azimuth direction $\phi_r = 0^\circ$ and a fixed codebook design. The phase shift follows the continuous phase shift design model.	20
5.2	IRS reflection gain (in dB) vs distance of the user from the IRS for a fixed codebook design with a radius of spherical grid at a position of 10 meters from the IRS and set of 0,1,2,3 bits for phase shift configuration.	21
5.3	The plot explains IRS reflection gain (in dB) against the evaluation angle θ_r when the user is at a fixed position at a distance of 30 meters with $\phi_r = 0^\circ$	23
5.4	IRS gain reflection (in dB) vs Evaluation Angles. For the simulation results we use the setting of PIN Diodes as bits and compare IRS reflection gains for the cases 1,2,3,10 bits and user at a distance of 30 meters in the direction of azimuth $\phi_r = 0^\circ$	24
5.5	In the figure we explore the averaged noise of the beamfocusing patterns for bits 1,2,3 and 10 for a fixed codebook design and user at a distance of 30 meters with $\phi = \{0, \pi\}$. However, we notice that as we go beyond 3 bits we come as close to our continuous model, which means 3 bits could be our a good tradeoff between hardware complexity and beamshape to determine the number of PIN Diodes we deploy.	26

Chapter 1

Introduction

The electromagnetic waves that carry information in wireless communications interact with objects and surfaces on their way from the transmitter to the receiver. Although the superposition of many propagation paths gives rise to random-like fading phenomena caused by a signal bouncing off the objects, every propagation path has a constant behavior. However, there are engineered materials whose interactions with electromagnetic waves are not constant but reconfigurable [3]. These materials are not naturally occurring but can be manufactured and deployed to shape the propagation environment [4, 3]. The prospects of including such Reconfigurable Intelligent Surfaces (RIS)'s as a part of beyond 5G network architectures are attracting much attention [2, 3]. RISs have also been called software-controlled metasurfaces [2] and intelligent reflecting surfaces (IRSs). When deployed in wireless networks, IRSs have the potential of turning the wireless propagation environment, which is highly probabilistic in nature, into a partially deterministic space, which is referred to as a smart (or intelligent) radio environment [4]

An IRS is a thin surface composed of N elements, each being a reconfigurable scatterer: a small antenna that reflects impinging signals without signal amplification while applying a phase shift to the reflected signal of the antenna elements. For narrowband signals, this delay between multipath signals corresponds to phase shifts [3]. Assuming the phase shifts are properly adjusted, the N scattered waves will add constructively at the receiver.

An IRS can be understood as a surface with real-time reconfigurable scattering properties that are controlled to improve communication performance. The unique feature of IRS is that it reduces the hardware complexity compared to half duplex Decode and Forward relaying schemes at the price of requiring a larger surface [2, 3]. In hindsight, IRS and relaying use cases seem very similar in theory with one of the major differences

being in terms of the almost passive behaviour of the IRS and the active behaviour of the relays. IRS functions opposite to the Decode and Forward relay case as there is no requirement of the IRS to decode the signals first but rather just reflect them. The figure shows an example use case scenario for an IRS where the Line-of-Sight (LoS) from the transmitter too the receiver is blocked and the communication relies on the IRS as a means to find a suitable path to the receiver.

In the example, we look at a scenario where the LoS path from the BS (transmitter) to the receiver (person) is blocked by the walls. The IRS on the wall close to the window provides a secondary path to the severely blocked communication system. The beam pattern we see is the reflected signals towards the receiver with the corresponding phase shift of the antenna elements.



Figure 1.1: Example use case scenario of IRS [3]

In the existing literature, the implementation of far-field phase shift codebook design has been extensively discussed [5, 2, 6]. However, for the far field models [5] the phase shift designs and beamforming requires the calculation of a large number of overheads (channel estimation pilots symbols for each propagation path to have reliable information for transmission creates large overhead and increases calculation for the same) during transmission. To mitigate the mentioned problem, the authors in [2, 5] proposed an implementation of offline codebook design to save the computational power and channel estimation overheads in the offline stage for the far field scenario. This essentially means the phase shifts are not calculated in an online manner which

introduced large overheads but we use the pre-defined offline codebooks which serve the transmission best [5, 7].

However, in our problem statement, we work focus on the mmWave frequency ranges which leads us to work in near field scenario because of such small wavelengths, and the effective IRS size is much larger with potentially hundreds of antenna elements [6]. That model the communication systems below sub 6 GHz frequencies which is the case for existing technologies. Hence, the effective near-field distance is much closer to the IRS. We further extend the linear codebook design in [2] to cater to the near-field case. And, designing such codebooks considering the impact of hardware impairments in the codewords namely phase noise and random noise.

In the next chapter, we introduce the extension of the Linear codebook design for satisfying near-field EM impinging wave gains for users near or close to the Bjornson distance and observe the differences between the theoretical and practical aspects of the near-field and far-field codebook designs. Furthermore, we will discuss how we calculate the phase shifts for the codebook design (beamfocusing) similar to match filtering or beamforming.

Chapter 2

Near and Far Field Distances for Extremely Large Antenna Arrays/IRS [1]

Since the introduction of Massive Multiple Input Multiple Output (MIMO) or Extremely Large Antenna Arrays (ELAA), the new generation of communication systems, has made the previous realization of systems with less number of antennas which functioned for far-field to function in the near-field scenarios. These ELAA-based communications system operates in Millimeter Wave (mmWave) or even THz frequencies range and hence employs hundreds or even thousands of antenna elements to satisfy the system requirements[8]. Therefore, the new systems are modeled in the near field scenario because the far field existed at a distance of hundreds of meters from the transmitter source ranges and thus the motivation for near field models.

The transmit antenna has an aperture that can be approximated as a continuum of point sources, each emitting a spherical wave. The combined wavefront can be approximated as a planar wave when observed at a sufficiently large distance. The corresponding region is called the far field and is characterized by the fact that the electric fields induced by the multiple point sources superimpose to create a field strength that is inversely proportional to the propagation distance d (valid also for near field). [9].

The near field case is defined when the transceiver is placed in proximity to the antenna array and the superposition of EM waves to the target is no longer compliant using planar waves but spherical waves [9]. These impinging spherical waves are thus defined using the information of the target's distance and angle in a 3D space.

From the literature [1, 9, 8], we know that the Fraunhofer distance, the classical way of defining the border between near-field and far-field, is a function of the antenna aperture size and the wavelength. However, in the paper [1], the author find out that for IRS with an extremely large number of antenna elements the border is defined using the antenna aperture and a number of antenna elements or as an IRS diagonal. We discuss the two definitions of antenna radiation field cases in the following sections.

2.1 Fraunhofer Distance and Fraunhofer Array Distance

To derive the Fraunhofer distance for the receiver antenna array and one transmitter, the following settings are used, an isotropic transmit antenna and a receiver with an aperture of size D as shown in the Figure 2.1 [9]. We consider the far field scenario where the superposition planar waves are simply and approximation and dependent on the angle towards the receiver and the phase variations are only dependent on the angle of arrival and departure not the distance.

Figure (2.1), shows the scenario where the transmitted wave impinges perpendicularly to the receiver in the x-y plane. If the propagation distance to the antenna's center is d , then the distance to the edges is $d' = \sqrt{d^2 + (D/2)^2}$. The additional distance the wave needs to travel to the receiver is given by $d - d'$, which corresponds to the maximum phase shift difference of the impinging waves between antenna elements. The phase shift difference is calculated as [1]:

$$\frac{2\pi(d - d')}{\lambda} = \frac{2\pi(\sqrt{d^2 + (\frac{D}{2})^2} - d)}{\lambda} \approx \frac{2\pi D^2}{\lambda 8d} \quad (2.1)$$

From the literature, we understand that the maximum phase shift that can be considered negligible is given as $\frac{\pi}{8}$ [9]. Thus, the Fraunhofer distance is given by replacing d as d_F in the above formula and equating to the phase shift of $\frac{\pi}{8}$.

$$\frac{\pi}{8} = \frac{2\pi D^2}{\lambda 8d_F} \quad (2.2)$$

and,

$$d_F = \frac{2D^2}{\lambda} \quad (2.3)$$

where, λ is the wavelength and d_F is the Fraunhofer distance.

Next, the authors in [1] define the Fraunhofer array distance for an ELAA which is

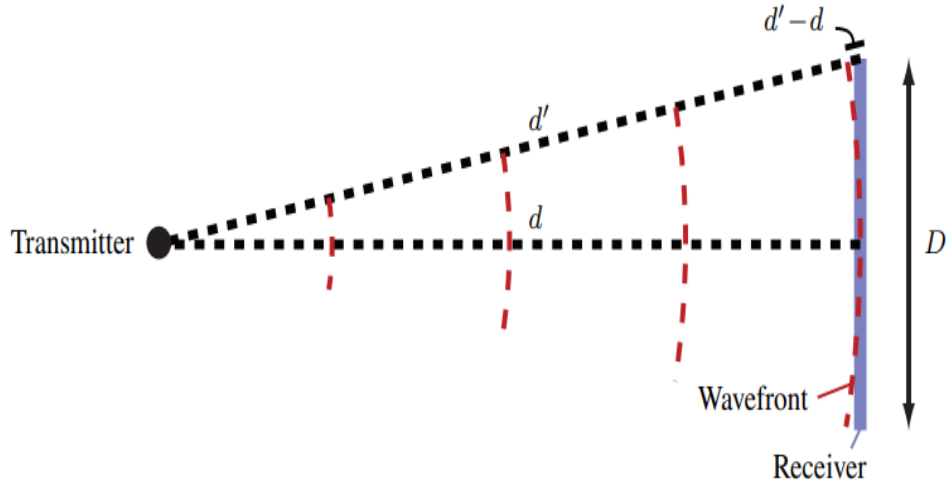


Figure 2.1: The phase shift between antenna elements is a result of impinging waves with a delay of $d - d'$ and the phase shift is given by $\frac{2\pi(d - d')}{\lambda}$

a planar array consisting of N identical antenna elements with the aperture area of each antenna as $A = \frac{\lambda^2}{4}$. Each unit cell or an antenna element has a diagonal D and $W = D\sqrt{N}$ is the maximum length (diagonal of the array) of ELAA. Replacing D by W in the above formula we approximate the Fraunhofer array distance [9]:

$$d_{FA} = \frac{2W^2}{\lambda} = \frac{2(D\sqrt{N})^2}{\lambda} = Nd_F \quad (2.4)$$

This means that the Fraunhofer array distance is N times the Fraunhofer distance.

2.2 Björnson Distance

The Fraunhofer distance is the measure of change of phase between antenna elements and is an important aspect to consider when considering far field cases, which assume phase shift error less than $\frac{\pi}{8}$ to be negligible. This assumption helps us in understanding far-field regime which then assumes approximately constant phase difference between antenna elements and acts as planar waves being reflected towards the receiver. However, the author in [1] explains the idea of managing reflected power towards the receiver, meaning providing the maximum signal strength possible to the receiver. This concept acts on providing max gain towards the receiver without taking into account the effect of phase differences between the antenna elements which we assume to be constant for uniform planar arrays. Accumulating all of this together defines the Björnson distance which essentially differentiates from which distance we gain the

most signal strength towards the receiver. In our project this particular definition is essential because our goal is to observe the behaviour of IRS gain amplitude not the phases.

The IRS reflection gain or ELAA gain when in the near field case saturates after performing beamforming or matched filtering (MF) [1]. This, however, occurs only after the Björnson Distance describes the border of near field and far field case for ELAA.

$$d_B = 2W = 2D\sqrt{N} \quad (2.5)$$

where, $W\sqrt{2NA}$ is the diagonal of the ELAA and $D = \sqrt{2A}$ is the diagonal of one antenna element. The expression here grows proportional to the $\sqrt{N}$ instead of N as in d_{FA} . This expression is independent of wavelength, meaning the phase shifts or EM wave phase delay to the receiver are not taken into account. Consequently, we have $d_{FA} \geq d_B$ and $d_B \geq d_F$ [1].

For the project, we consider the IRS of size (aperture size) $(10\lambda, 10\lambda)$, and each antenna element is isotropic and identical by nature with each unit cell having uniform spacing of (d_x, d_y) . The number of IRS elements is $N=400$, with antenna aperture $A = \frac{\lambda}{2}$. The normalized Björnson distance over wavelength for this case is $d_B = 28.28$ and the normalized Fraunhofer array distance over wavelength is $d_{FA} = 400$. Björnson distance typically relates to the fact that about 96% of the total IRS reflection gain is achieved [1].

Chapter 3

System Model: Far-Field Scenario

In the system model, we present the IRS unit cell phase shifts using the phase shifts model which is designed heuristically the unit cell phase shift for every transmission scenario and transmission model where the pre-defined set of phase shifts are used for transmission in the far field scenario. We first present the system model equations for the two modes consisting of links from BS to IRS, IRS to the receiver, and BS to the receiver [5]. Further, we discuss this in the later section of the codebook designs [2] and set our reference codebook to Linear Model for far-field scenarios and use IRS response gain equations with Linear codebook design.

In the next chapters, we will discuss our preliminary system model for the base far-field model and observe if the codebook designs for far-field model can be employed for the near-field case. EM wave function characteristics lead us to work on dealing with far-field and near-field models separately and differentiate between these borders as use case scenarios for IRS.

3.1 System Model

Let us consider an example downlink network (case with SISO with IRS assisted communication) consisting of a base station (BS) equipped with $N_t = 1$ transmit antennas serving $k = 1$ mobile users each equipped with $N_r = 1$ receive antennas. The direct links between the BS and the users may exist but are severely shadowed (e.g., by a blocking building). The downlink communication is assisted by an IRS that comprises $Q = Q_x Q_y$, Q_x and Q_y are the IRS elements along the x and y axis. The end-to-end system model can be written as [2, 5]:

$$\mathbf{y}_k = (\mathbf{H}_{d,k} + \mathbf{H}_{r,k} \mathbf{\Omega} \mathbf{H}_i) \mathbf{x}_k + \mathbf{n}_k \quad (3.1)$$

where: $\mathbf{y}_k \in \mathbb{C}^{N_r}$ is the output at the receiver, $\mathbf{H}_{d,k} \in \mathbb{C}^{N_r \times N_t}$ is the channel from BS-user, $\mathbf{H}_{r,k} \in \mathbb{C}^{N_r \times Q}$ channel vector from IRS-user, $\Omega \in \mathbb{C}^{Q \times Q}$ is the diagonal IRS phase shift matrix for user k, and $\mathbf{H}_{i,k} \in \mathbb{C}^{Q \times N_t}$ is the channel vector from BS-IRS, $\mathbf{x}_k \in \mathbb{C}^{N_t}$, and $\mathbf{n}_k \in \mathbb{C}^{N_r}$ are the transmitted signal and the noise at user k. The codebook Ω is defined as :

$$\Omega = [\bar{g}e^{jw^1}, \dots, \bar{g}e^{jw^Q}] \quad (3.2)$$

3.2 Transmission Mode

The transmission-mode model introduced in [2] relies on designing a predefined set of M transmission modes configurations for codebooks in an offline stage which later can be selected for online transmission. The IRS response function of transmission mode m is given by $g_m(\Psi_r, \Psi_i)$ denote the normalized IRS response function for the m-transmission mode, where $\Psi_i = (\theta_i, \phi_i)$, and $\Psi_r = (\theta_r, \phi_r)$ are the Angle of Arrival (AoA) of the incident wave and the Angle of Departure (AoD) of the reflected wave, respectively, and θ and ϕ re-used to denote the elevation and azimuth angles, respectively. Now we define a signal model equation derived in [2, 5] to calculate IRS response function $g_m(\Psi_r, \Psi_i)$ gain for far field and later manipulate to fit the near field scenario.

$$\mathbf{y}_k = (\mathbf{A}_{d,k} \Sigma_{d,k} \mathbf{D}_{d,k}^H + \mathbf{A}_{r,k} \Sigma_{r,k} \mathbf{G}_k \Sigma_i \mathbf{D}_i^H) \mathbf{x}_k + \mathbf{n}_k \quad (3.3)$$

where $\mathbf{G}_k \in \mathbb{C}^{L_r \times L_i}$, L_r and L_i are the multipath components of IRS to Receiver and BS to IRS respectively. And, $(A_{d,k}, A_{r,k})$ is the steering vector for the angle of arrival for user k and links BS to IRS and IRS to the receiver. Similarly, $(D_{d,k}, D_{r,d})$ the transmit steering matrix for the angle of departures for user k and links BS to IRS and IRS to the receiver. The decomposition matrices $\Sigma_{d,k}$ and $\Sigma_{d,k}$ are the channel gain matrices for the links.

The IRS gain matrix $\mathbf{G}_k$ is given by [2]:

$$\mathbf{G}_k = \mathbf{D}_{r,k}^H \Omega \mathbf{A}_i = \sum_{m=1}^M s_m \mathbf{G}_{m,k} \quad (3.4)$$

where, the n_r -th row and n_i -th column of $\mathbf{G}_{m,k} = \mathbf{D}_{r,k}^H \Omega_m \mathbf{A}_i$ is the IRS response function $g_m(\Psi_i, \Psi_r)$ evaluated at n_i -th AoA of the IRS and n_r -th AoD from the IRS for user k. The IRS response gain for the link IRS to receiver for transmission mode m is derived from equation (3.2) is given by:

$$g_m(\Psi_i, \Psi_r) = \mathbf{d}_{r,k}^H \Omega_m \mathbf{a}_i \quad (3.5)$$

$\mathbf{a}_i$ is the steering vector for the angle of arrival and $\mathbf{d}_{r,k}$ is the transmit steering matrix for the angle of departure [2].

IRS acts as an anomalous scatterer as the phase shift configurations direct the reflected signals through multiple channels in the wireless medium toward the user [5]. We know in the far field the curvature of the wavefront originating from a distant source acts as a planar wavefront while it becomes a spherical wavefront in the near field scenario. IRS achieves its objective just by making use of phase shifts in the codebook design of IRS as a beamformer in the case of the far-field case and as a beamfocusing in the near field case.[8, 6]

In the next sections, the codebook designs in [2] are discussed, and in this project, we first consider baseline Linear phase shift design and how IRS response gains are modeled for first the far field and then the near field for the respectively.

3.3 DFT Phase Shift Design

DFT-based phase shift codebook design for the IRS elements is the baseline method to design unit cell elements of the IRS as described in [2]. Phase shift applied for the unit cell at (n_x, n_y) -th position is given by [2]:

$$e^{jw_{n_x, n_y}} = e^{-jkm_x n_x} e^{-jkm_y n_y} \quad (3.6)$$

where, w_{n_x, n_y} is the (n_x, n_y) -th codeword of the IRS unit cell. And, wave number $k = \frac{2\pi}{Q}$.

The IRS response gain for AoA and AoD in the far field case is defined as $g_m(\Psi_i, \Psi_r)$ [2, 5] and expanded below as :

$$g_m(\Psi_i, \Psi_r) = \bar{g} \sum_{n_x=0}^{Q-1} \sum_{n_y=0}^{Q-1} e^{\frac{j2\pi k A_x(\Psi_i, \Psi_r) d_x}{\lambda} n_x} e^{\frac{j2\pi k A_y(\Psi_i, \Psi_r) d_y}{\lambda} n_y} e^{jw_{n_x, n_y}} \quad (3.7)$$

where: $A_x(\Psi_i, \Psi_r)$ and $A_y(\Psi_i, \Psi_r)$ are the steering angle with AoA and AoD respectively. Angle of arrival and respectively defined by, $A_x(\Psi_i, \Psi_r) = A_x(\Psi_i) + A_x(\Psi_r)$ with, $A_x(\Psi) = \sin \theta \cos \phi$ and $A_y(\Psi) = \sin \theta \sin \phi$.

The steering vectors from equation (3.5) for AoA Ψ_i and AoD Ψ_r are given by $e^{\frac{j2\pi(d_x A_x(\Psi_i) n_x + d_y A_y(\Psi_i) n_y)}{\lambda}}$ and $e^{\frac{-j2\pi(d_x A_x(\Psi_r) n_x + d_y A_y(\Psi_r) n_y)}{\lambda}}$. k is the wave number and unit cell spacing (d_x, d_y) . Each

unit cell spacing is given by $d_x = \frac{\lambda}{2}$.

Here, the setting considered for later examples is $\Psi_i = (\theta_i, \phi_i)$ as a pair $(0^\circ, 0^\circ)$. $\phi_r = (0, \pi)$, and $\theta_r = [0, \frac{\pi}{2}]$ for $\phi_r = (0, \pi)$.

3.4 Linear Phase Shift Design[2]

The phase shift of the codebook design here is a linear functions with constant slopes of $\frac{2\pi m_x}{Q_x}$ and $\frac{2\pi m_y}{Q_y}$ at points (n_x, n_y) -th unit cell position on the IRS. In the linear phase shift design, phase shifts are similar to the DFT-based design with a slight change in the total codebook size M [2]. We consider here the $M = M_x M_y$ where M is the codebook size corresponding to our requirement of fewer beams to sweep the users for a AoD given an AoA. The phase shift model is given by [5]:

$$e^{jw_{n_x, n_y}} = e^{(-j\frac{2\pi d_x \bar{B}_x}{M_x \lambda} m_x n_x)} e^{(-j\frac{2\pi d_y \bar{B}_y}{M_y \lambda} m_y n_y)} \quad (3.8)$$

where: $m_t = 0 \dots M_t - 1$, $t \in \{x, y\}$ is the (m_x, m_y) pair of $m = 0, \dots, M - 1$ for $M = M_x M_y$, and codebooks for the (n_x, n_y) pair of IRS elements. $\bar{B}_t$, $t \in \{x, y\}$ is the range of phase shift gradients. And, phase gradient is defined as in [2], $\bar{B}_t = \min\{4, \frac{\lambda}{d_t}\}$. This linear phase shift design gives us the desired power efficiency for the reflected EM wave while reducing the number of codebook elements.

IRS response function for a codebook element $m \in (m_x, m_y)$ -th transmission mode is again given using the equations (3.5 and 3.7).

where: g_m is the IRS response gain for codebook element m . $d_r^H \in \mathbb{C}^Q$, $\Omega \in \mathbb{C}^{Q \times Q}$ and $a_i \in \mathbb{C}^Q$ are the angle of departure vector, phase shift codebook and angle of arrival vector, respectively.

3.5 Quadratic Phase Shift Design[2]

To make IRS codewords even less in number and broaden the beamwidth required to cover the user space, in other words less number of beams to cover user space since the calculation of such large number of codewords increases computational complexity and hardware requirements, the concept of quadratic phase shift design was introduced [2]. Extending the linear phase shift design the authors in [2] define phase shifts as:

$$e^{jw_{n_x, n_y}} = e^{\frac{-j2\pi d_x}{\lambda} \left[\frac{\Delta \bar{B}_{x, m_x} n_x^2}{2Q_x} + \bar{B}_{x, m_x} n_x \right]} e^{\left[\frac{-j2\pi d_y}{\lambda} \left(\frac{2\pi \Delta \bar{B}_{y, m_y} n_y^2}{2Q_y} + \bar{B}_{y, m_y} n_y \right) \right]} \quad (3.9)$$

where: $B_{x,m_t}^- \in [0, \bar{B}_t]$, and $\bar{B}_t = \min\{4, \frac{\lambda}{d_t}\}$, $d_t = \lambda/2$ and $t \in \{x, y\}$. We define uniform quantization of $\Delta B_{t,m_t} = \frac{\bar{B}_t}{M_t}$. And, the update rule for phase shift gradient $B_{t,m_{t+1}}^- = m_t \Delta B_{t,m_t}$, for $m_t = 0, 1, \dots, M_t$.

In chapter 4, we define the IRS response gain for the continuous phase shift model using the Linear codebook design from chapter 3 section 2. We discern that the Linear codebook design for far-field case with evaluation direction lying in front of the IRS along the receiver direction cannot be used as a phase shift design scheme as the wavefront for near field is not negligible but spherical. Therefore, we define a new codebook or phase shift design that conforms to the near-field regime modeling reflected Electromagnetic Waves (EM) waves as spherical waves.

Chapter 4

Uniform 3D Grid Codebook Design for Near Field

After discussing the two different definitions of far field and near field which differentiates between far field and near field scenarios we move our focus towards modeling codebooks for near field scenarios and mmWave frequency range communications system. Here, presented is the beam management framework for mmWave multi-antenna communication system. We design a fixed phase shift codebook for near-field communication scenarios, in this case, the user lies in the same plane as the EM wave propagation direction. The idea is to configure the phase shift as matched filters to have a directed gain across specific frequency. We need to consider the fact that the EM wave in the near field case is modeled as a spherical wave that employs the information of the user's position (distance, angle) to evaluate the EM gain [8, 6].

4.1 System Model: Near-field scenario

Capitalizing on the near field hierarchical IR response gain in [6], we devise an IRS gain profile and RIS to BS phase shifts. Our goal is to directly use the formula for near-field IRS reflection gain in [6] and use it as a reference for our phase shift design and compare it with the formerly used linear codebook design for the far-field scenario. We define the near field response gain formula and in the later sections use it to model and design our proposed phase-shift model.

We define the Base station (BS) as a point source transmitter with LOS components from BS to RIS and BS to the receiver. The IRS response function is given by [6]:

$$g_m(\mathbf{p}_i, \mathbf{p}_r) = \bar{g} \sum_{n=0}^{Q-1} e^{\frac{j2\pi}{\lambda} [\|\mathbf{p}_i - \mathbf{p}_n\| + \|\mathbf{p}_r - \mathbf{p}_n\|]} e^{jw_n} \quad (4.1)$$

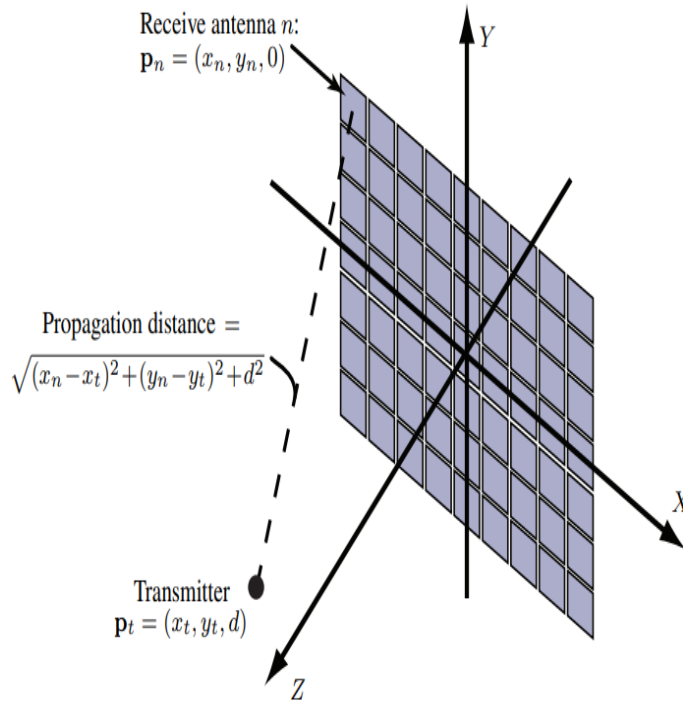


Figure 4.1: The figure shows the IRS lies in the xy plane and the direction of the wave propagation. The point source transmitter lies at a position of $[0, 0, d]$, where d is the distance from the IRS center to the BS [3]

where: $\mathbf{p}_i$, $\mathbf{p}_r$, $\mathbf{p}_n$ are the position of the BS, position of the target or receiver, and n^{th} IRS element. The math symbol $||\cdot||$ refers to the L2 norm or magnitude of the vector. w_n is the phase shift for the n^{th} IRS element which in this model we take from previously defined linear codebook design in chapter 3 and calculate gain based on the new formulation [6].

$\mathbf{p}_i = [x_{BS}, y_{BS}, z_{BS}]$ is the transmitter position. In the near-field scenario, we must focus on BS-RIS and RIS-MU LOS paths, and here RIS aims to illuminate the receiver at a target user position $\mathbf{p}_r$.

$$\mathbf{p}_r = [x_{rx}, y_{rx}, z_{rx}] \quad (4.2)$$

x-coordinate of $\mathbf{p}_r$ is defined as $(d \sin \theta_r \cos \phi_r)$, y-coordinate is defined as $d \sin \theta_r \sin \phi_r$, and z is $(d \cos \theta_r)$, where, (θ_r, ϕ_r) are the elevation and azimuth angles in the same direction as in the reflected wave. The evaluation direction for the reflected EM wave towards the receiver is in the xz plane for the full range of angel sweep $[\frac{-\pi}{2}, \frac{\pi}{2}]$. And, $\mathbf{p}_b$ is the center of the observation area which is optional meaning it could be either origin or arbitrary coordinates. Q_x and Q_y are the number of RIS elements in x and y directions.

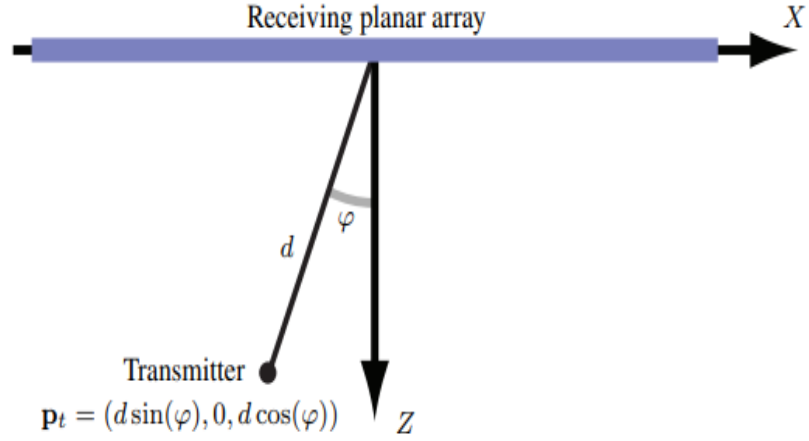


Figure 4.2: Receivers position with the transmitter in the figure acting as a receiver for our downlink case and IRS is the transmitter which reflects the signal towards the receiver [1]

We formulate the logic to define $\mathbf{p}_n$ vectors as:

$$\mathbf{p}_n = [x_0 + md_x/2, y_0 + nd_y/2, 0] \quad (4.3)$$

where y_0 and x_0 are the origin coordinates of the RIS and (d_x, d_y) are the RIS elements spacing factors, and m, n are the index of the RIS elements.

4.2 Phase Shifts Design using Uniform 3D Grid Space

We continue our discussion from the previous section about near-field codebook design using spherical wave characteristics such as distance and angle to define the transmitter and receiver positions. To define our proposed phase shift design, we consider two important aspects,

- we will use spherical coordinates to model reflected waves as the EM wave wavefront does not follow the planar wave model anymore.
- We make use of the coordinates in the spherical system to define our uniform 3D grid and also the receiver position.

Now, we employ near-field beamforming design in a way that would follow phase shifts to act similar to match filters at the receiver front. The phase shift design would follow, the formula:

$$w_n = \frac{-2\pi(||\mathbf{p}_i - \mathbf{p}_n|| + ||\mathbf{p}_t - \mathbf{p}_n||)}{\lambda} \quad (4.4)$$

Now we define, the $\mathbf{p}_t$ vectors in spherical co-ordinates as,

$$p_t = [r \sin(\frac{m_x \pi}{M_{xz}}) \cos(\frac{m_x \pi}{M_{xz}}), r \sin(\frac{m_y \pi}{M_{yz}}) \sin(\frac{m_y \pi}{M_{yz}}), r \cos(\frac{m_z \pi}{M_{yz}})] \quad (4.5)$$

where r is the radius of the various concentric spherical grids which act as a wave-front, that move away from the IRS as a function of wavelength. $m_t = 1, 2, \dots, M_t$ for $t \in (x, y, z)$ are the discrete points on the spherical grid to emulate the target positions.

The goal of phase shift design is to essentially focus the reflected signal from the IRS to the receiver with the peak gain or signal strength being highest at the position where both $\mathbf{p}_t$ and $\mathbf{p}_r$ overlap.

4.3 Results

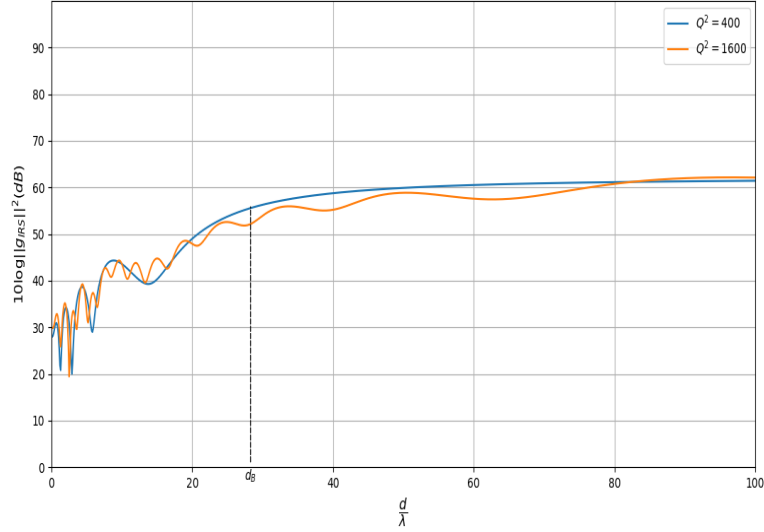
In this section, we discuss and compare far-field and near-field IRS response gains against the distance of the user from the IRS center. The settings used for the simulation results are as follows: The position vector of the point source BS is $\mathbf{p}_i = [0, 0, d = 1000] * \lambda$, where $\lambda = 0.01\text{m}$ is the wavelength corresponding to the frequency of 30GHz $\mathbf{p}_r$ and $\mathbf{p}_t$ are functions of $[r, \theta_r = 0^\circ, \phi_r = 0^\circ]$ which is similar to setting used in [2], also mentioned in chapter 3, the radius of the spherical grid takes on values from the set [10, 20, 30, 40, 60, 100] meters. The IRS position vectors are defined as above in section 4.1.

In the following, we compare far-field and near-field case using Linear Codebook and our proposed phase shift design, respectively [10]. The former was studied extensively in the literature, however, the codebook design for near-field scenarios is not precisely defined but stated and studied, and here we present a phase shift design to ultimately maximize the IRS reflection gain by using beamforming or match filtering.

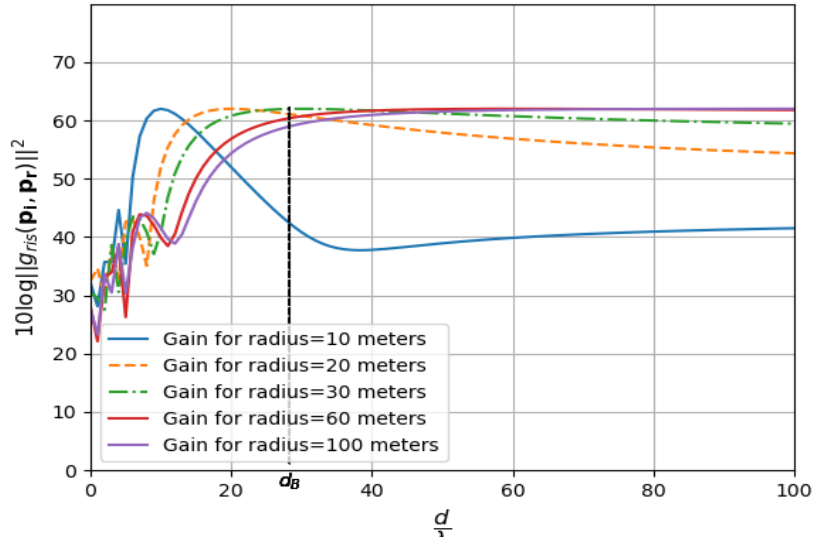
In the following figures, firstly, we interestingly note that the use of Linear codebook design for far-field and how the number of IRS elements affects the IRS gain. Secondly, we note the use of our new phase shift design in the gain formulation from [6] works in the near field regime. We establish the fact in chapter 2 that the codebook design for far-field cannot be used in the near-field scenarios. Therefore, we employ the use of our proposed codebook design model for the near-field case. The user is at a position on the spherical grid and the IRS response gain is calculated for various distances of the user from the IRS. In the figure, we use distances of 10, 20, 30, 60 and

100 meters for defining a spherical grid and 1000 realizations of distance from the IRS for one fixed codebook model.

The IRS response gain in Fig (b) is the highest with the peak at the exact receiver position for each codebook design (at a radius for spherical grid design) and saturates after Bjornson distance (d_B).



(a)



(b)

Figure 4.3: In Fig (a), IRS response function gain (in dB) vs Distance from the center of the IRS, the Linear codebook for the far-field case using the near-field gain formula [2]. In Fig (b), we use our proposed phase shift design for near-field scenarios and use the near-field gain formula. We plot IRS response gain for various spherical grids, with a constant codebook, and the plot also explains the user position on the spherical grid at radius

Chapter 5

Discrete Phase Shift design and Effects of Erroneous Phase Shift Settings

In this chapter, we move forward with modeling discrete phase shift design from our proposed continuous phase shift based Extension of Linear Codebook design. We also discuss the effects of noise and phase errors in the discrete phase shift model and conclude the codebook design regime for near-field scenario.

In the previous chapters and discussions of Linear Codebook Design for IRS in the offline stage to reduce the number of phase shift calculations, we considered a fixed phase shift designs to reduce the large overheads because of channel estimation. However, in this case, because of the limited number of phase shifts considered for IRS reflections gain towards the desired user we observe some loss of reflections gains because of inaccuracies introduced while considering only a set of phase shifts.

These limited number of phase shifts fail to provide the exact reflection gains as the continuous Linear codebook phase shifts i.e real numbers as phase shifts from $[0, 2\pi]$ and therefore, we observe the loss of IRS response function gain. The tradeoff between the limited number of phase shift configurations used and IRS reflection gain is required to mitigate the use of resources such as power and computing complexity [2, 3].

Accordingly, we implement the predefined set of discrete phase shifts as an example case in: $w_n = \{0, 80^\circ, 160^\circ, 260^\circ\}$.

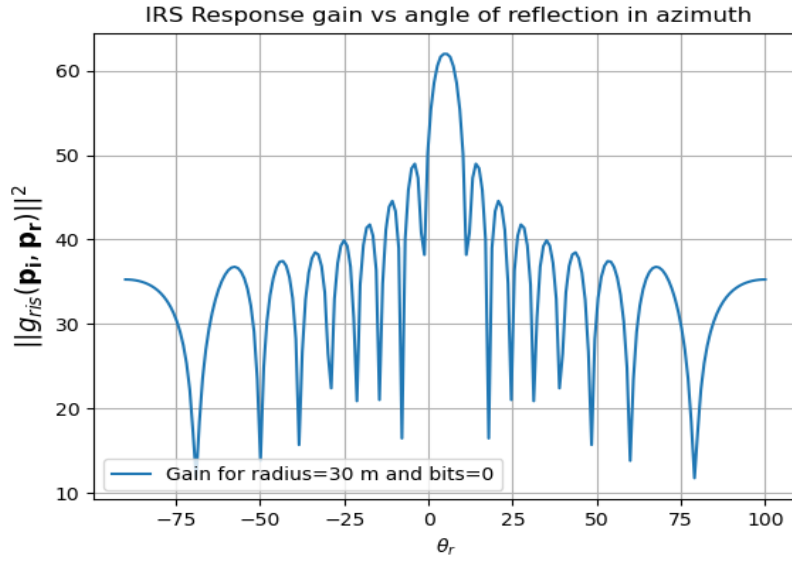


Figure 5.1: IRS reflection gain (in dB) against evaluation angles θ_r for a fixed user at a distance of 30 meters, azimuth direction $\phi_r = 0^\circ$ and a fixed codebook design. The phase shift follows the continuous phase shift design model.

The following sub-sections consist of figures and an explanation of the IRS response gain function with discrete phase shift designs and a comparison with the previous continuous phase shift model.

5.1 Discrete Phase Shift Model

In this section, as stated before, we look to model a codebook using a pre-defined set of discrete phase shifts. As the IRS elements are controlled using PIN Diodes which can only configure phase shifts in a discrete fashion, namely, using the number of PIN Diodes or Bits to configure phase shifts; for example 2 bits corresponds to a phase shift of $\{0^\circ, 80^\circ, 160^\circ, 260^\circ\}$. We consider the choice of phase shifts from the discrete set of phase shifts according to the number of PIN Diodes, we used to generate the phase shifts and call them as bits. The term bits, is used to select the phase shifts from the defined set according to the equation $mul = \frac{2\pi}{2^b}$ where b =bits (number of PIN Diodes) and the variable mul implies the multiplication factor for phase shift discretization. We use the terms bits and PIN Diodes interchangeably.

The continuous phase shift model, it creates a hardware limitation in terms of installing such a large number of PIN Diodes to approximate continuous phase shifts. Thus, we use a set of discrete phase shifts to approximate the IRS gain as in the contin-

uous phase shift model.

In the Figure 5.2, we observe the IRS reflection gain for the fixed user position at a distance of 10 m from the IRS using a fixed codebook. From the figure, we can observe that we can get the closest to the original reference figure 4.1 where no bits were used (continuous fixed phase shift model), only using the 3 PIN Diodes.

The figure also shows how the IRS reflection gain saturates after the Bjornson distance ($d_B = 2D\sqrt{N}$, where $D = \sqrt{2A}$, A is the antenna element aperture size $A = \frac{\lambda^2}{4}$, and N is the number of antenna elements) at 28.28 m (derived in chapter 2). The reflection gain is at the highest for the fixed user at 10 m meaning the IRS elements focus the reflections at the user at 10 m and subsequently lower the reflection gain as we move further away.

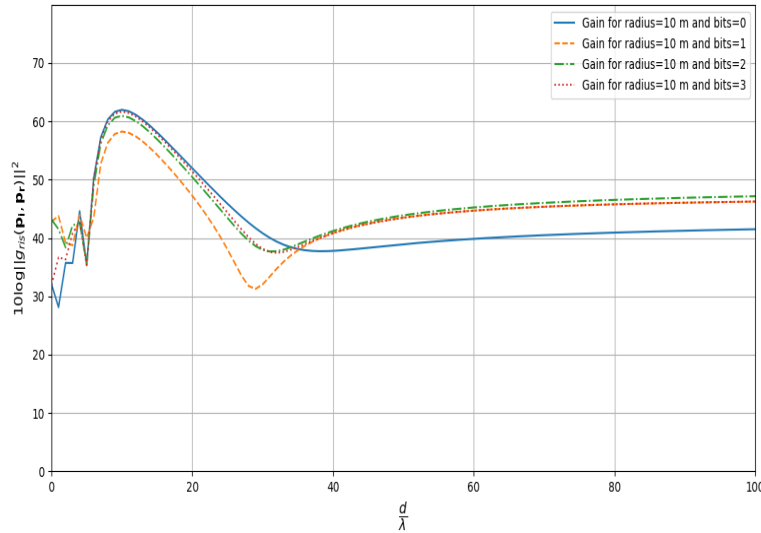


Figure 5.2: IRS reflection gain (in dB) vs distance of the user from the IRS for a fixed codebook design with a radius of spherical grid at a position of 10 meters from the IRS and set of 0,1,2,3 bits for phase shift configuration.

5.2 Effect of Phase Errors and Gaussian Noise on Linear codebook phase shifts

In this section, we discuss how other hardware impairments such as phase errors and gaussian noise affect the beamshape and essentially the IRS reflection gain. The

phase errors in IRS antenna elements caused by hardware errors or erroneous calculation of codebook phase shifts lead to inaccurate IRS reflection gain and user localization.

Next, we observe how noises such as gaussian noise caused by PIN Diodes circuitry [3] affects the phase shifts of IRS antenna elements and also the IRS reflection gains when evaluated for evaluation angle θ_r in the elevation angle set $[0^\circ, 90^\circ]$ in the figure 5.5.

Here, we consider our discrete phase shift model to stay in line with the hardware impairments. Also, to further study the effect of these impairments and errors on the IRS response gain function we average the effect of errors over several realizations of IRS response gain.

5.2.1 Effect of phase errors on IRS beam pattern

In this sub-section, as stated in the above section we study the effect of phase errors in the range of +10 degrees to -10 degrees on beamforming for the near-field scenario. The effect of phase errors in the linear codebook design is the reduction in the IRS reflection gain as compared to the previously used continuous phase shifts.

In figure 5.3, the impact of phase errors (one realization of the result) as an example for realizing the figure between $[\frac{-\pi}{12}, \frac{-\pi}{12}]$, reduces the IRS response gain along with visible distortions in the beam pattern along the evaluation angle θ_r . The evaluation direction for this one realization of the effect of phase errors in the codebook in range of $[0^\circ, 90^\circ]$ with a fixed target position at 30 meters for a fixed codebook design in the direction of $\phi_r = 0^\circ$.

In figure 5.3 , we average the IRS reflection gain over the phase errors for 100 realizations and observe that the peak of the IRS reflection gain and shape of the main lobe of the beam pattern is slightly different (distorted at the extreme sides) than from figure 5.1. However, we get smoother side lobes by averaging the IRS reflection gain beam pattern for 100 realizations. The same settings have been used for figure 5.3 as in figure 5.1.

As we move towards the extreme angles of the evaluation angle set we see more than one peak which explains inaccurate beamforming for the user at that position which essentially explains the effect of phase errors in the codebook phase shifts.

In the following figures 5.4 a) and b), we discuss the beam pattern of the fixed codebook design for a fixed user position at $\theta = 0^\circ$ and $\phi = 0^\circ$, and at a distance of

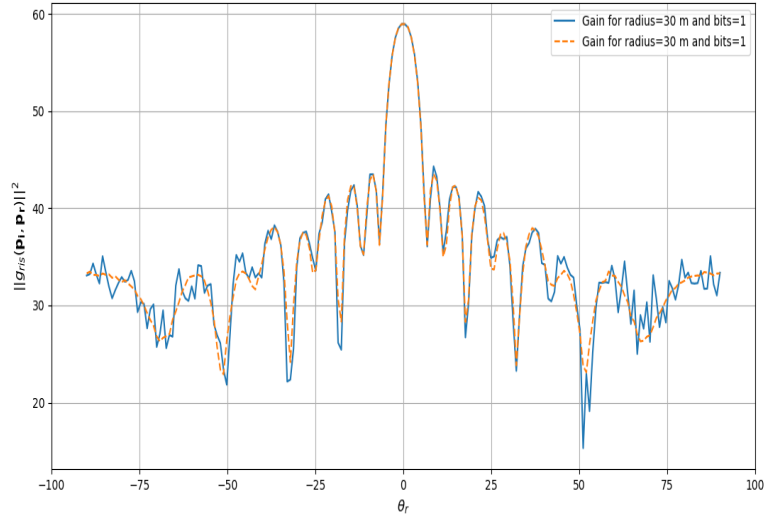
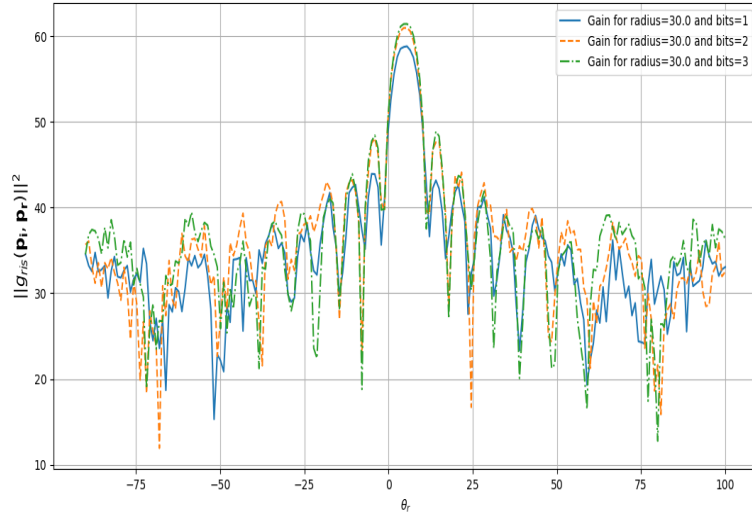


Figure 5.3: The plot explains IRS reflection gain (in dB) against the evaluation angle θ_r when the user is at a fixed position at a distance of 30 meters with $\phi_r = 0^\circ$.

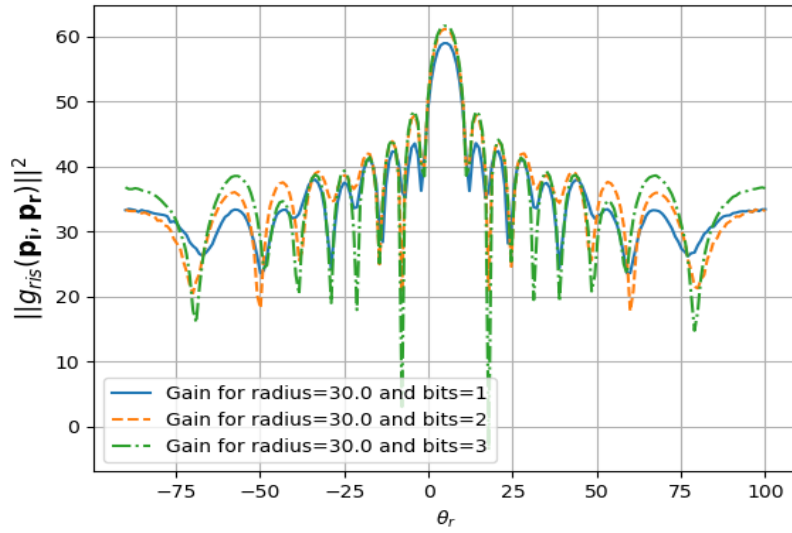
30 meters against the evaluation angle. The main lobe of the beam pattern shows the position of the fixed user.

In figure 5.4 (a), we plot the IRS reflection gain against the evaluation angle over one realization for the discrete phase shift model with phase shift errors. The plot shows that for evaluation angles at the extreme ends, the IRS reflection gain consists of distorted side lobes, which means inaccurate beamforming at that position and loss of gains as we move away from the user position.

In figure 5.4 (b), we plot the average IRS response gain for 100 realizations of phase errors and get averaged out IRS reflection gain and focused side lobe gain levels. Henceforth, we deploy the case of 10 PIN Diodes as a higher limit which is very close to the continuous phase shift model. Also, we see how the choice of discrete phase shifts affects the beam pattern for a user at a fixed distance of 30 meters. And we then infer essentially how many PIN Diodes we need to model our IRS phase shifts close to the initial theoretical codebook design model.



(a)



(b)

Figure 5.4: IRS gain reflection (in dB) vs Evaluation Angles. For the simulation results we use the setting of PIN Diodes as bits and compare IRS reflection gains for the cases 1,2,3,10 bits and user at a distance of 30 meters in the direction of azimuth $\phi_r = 0^\circ$

5.2.2 Impact of Gaussian Noise on Fixed Codebook Phase Shifts

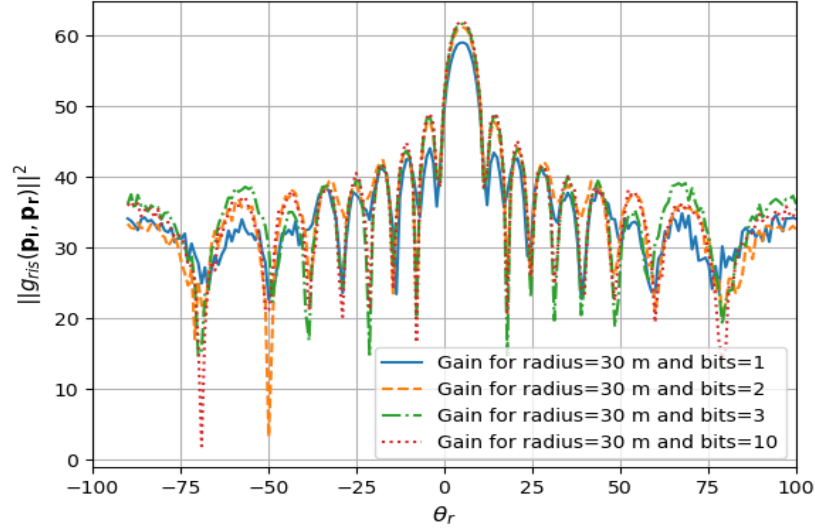
In this section, we infer the effect of gaussian noise on the phase shifts of the IRS codebook and the IRS response gain. The equation for phase shifts with additive white gaussian noise is given by [11]:

$$g_m(\mathbf{p}_i, \mathbf{p}_r) = g \sum_{n=0}^{Q-1} e^{\frac{j2\pi}{\lambda} (\|\mathbf{p}_i - \mathbf{p}_n\| + \|\mathbf{p}_r - \mathbf{p}_n\|)} e^{-j(w_n + \phi_n)} \quad (5.1)$$

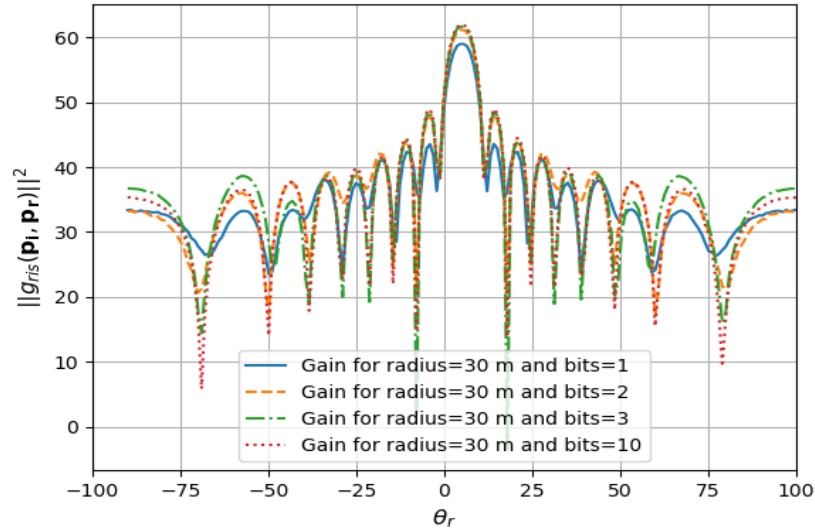
where, $\phi \in \mathcal{C}^{Q \times Q}$ is the gaussian noise matrix and ϕ_n is an (n_x, n_y) -th element of the noise matrix.

In the following figures 5.5 a) and b), we discuss the beam pattern. The main lobe of the beam pattern helps in discerning the position of the desired user. Also, we see how the choice of discrete phase shifts affects the beam pattern for a given user at a distance of 30 meters in the direction with azimuth angle $\phi = \{0^\circ, \pi\}$.

The simulation results explain the effect of gaussian noise on the IRS's phase shifts over one realization of gaussian noise (figure 5.5 (a))and over 100 realizations of gaussian noise (figure 5.5 (b)). In figure 5.5 (b), we see the averaged effect of gaussian noise on the IRS elements. The case with 10 bits implies that beyond 3 PIN Diodes the beam shape close to the continuous codebook model as defined in the previous sections.



(a)



(b)

Figure 5.5: In the figure we explore the averaged noise of the beamfocusing patterns for bits 1,2,3 and 10 for a fixed codebook design and user at a distance of 30 meters with $\phi = \{0, \pi\}$. However, we notice that as we go beyond 3 bits we come as close to our continuous model, which means 3 bits could be our a good tradeoff between hardware complexity and beamshape to determine the number of PIN Diodes we deploy.

Chapter 6

Conclusion

In the project we discuss how the extensively studied far-field codebook design in the literature does not conform to the near field gain formulation and we need to design a near field codebook design to conform to the near field gain equation. Since the Linear codebook design for far-field case does not provide beamfocusing in the near-field regime we are convinced to adapt to other design which is not modeled using planar wave characteristics but rather employ spherical wave characteristics. Then we compare the two models and to limit the use of hardware resources we employ discrete phase shift designs.

Bibliography

- [1] Parisa Ramezani and Emil Björnson. Near-field beamforming and multiplexing using extremely large aperture arrays. *arXiv preprint arXiv:2209.03082*, 2022.
- [2] Vahid Jamali, Marzieh Najafi, Robert Schober, and H Vincent Poor. Power efficiency, overhead, and complexity tradeoff of IRS codebook design—Quadratic phase-shift profile. *IEEE Communications Letters*, 25(6):2048–2052, 2021.
- [3] Emil Björnson, Henk Wymeersch, Bho Matthiesen, Petar Popovski, Luca Sanguinetti, and Elisabeth de Carvalho. Reconfigurable intelligent surfaces: A signal processing perspective with wireless applications. *IEEE Signal Processing Magazine*, 39(2):135–158, 2022.
- [4] Marco Di Renzo, Konstantinos Ntontin, Jian Song, Fadil H Danufane, Xuewen Qian, Fotis Lazarakis, Julien De Rosny, Dinh-Thuy Phan-Huy, Osvaldo Simeone, Rui Zhang, et al. Reconfigurable intelligent surfaces vs. relaying: Differences, similarities, and performance comparison. *IEEE Open Journal of the Communications Society*, 1:798–807, 2020.
- [5] Marzieh Najafi, Vahid Jamali, Robert Schober, and H Vincent Poor. Physics-based modeling and scalable optimization of large intelligent reflecting surfaces. *IEEE Transactions on Communications*, 69(4):2673–2691, 2020.
- [6] George C Alexandropoulos, Vahid Jamali, Robert Schober, and H Vincent Poor. Near-Field Hierarchical Beam Management for RIS-Enabled Millimeter Wave Multi-Antenna Systems. *arXiv preprint arXiv:2203.15557*, 2022.
- [7] Walid R Ghanem, Vahid Jamali, Malte Schellmann, Hanwen Cao, Joseph Eichinger, and Robert Schober. Optimization-based Phase-shift Codebook Design for Large IRSs. *arXiv preprint arXiv:2203.01630*, 2022.
- [8] Mingyao Cui, Zidong Wu, Yu Lu, Xiuhong Wei, and Linglong Dai. Near-field communications for 6G: Fundamentals, challenges, potentials, and future directions. *arXiv preprint arXiv:2203.16318*, 2022.

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- [9] Krishnasamy T Selvan and Ramakrishna Janaswamy. Fraunhofer and Fresnel Distances: Unified derivation for aperture antennas. *IEEE Antennas and Propagation Magazine*, 59(4):12–15, 2017.
 - [10] Xiang Li, Xin Wang, Xiaolin Hou, Lan Chen, and Satoshi Suyama. Two-Step Beamforming Scheme for Large-Dimension Reconfigurable Intelligent Surface. In *2022 IEEE 95th Vehicular Technology Conference: (VTC2022-Spring)*, pages 1–5. IEEE, 2022.
 - [11] Thomas Höhne and Ville Ranki. Phase noise in beamforming. *IEEE Transactions on Wireless Communications*, 9(12):3682–3689, 2010.